

Solution Requirements Document

VANGUARD CREDIT AUTOMATION | ACADEMIC PROJECT TRACK DELIVERABLE

1. Functional Requirements

- FR-1: Pipeline must evaluate Logistic Regression, Decision Tree, Random Forest, and XGBoost.
- FR-2: Active model selection must be based on Highest Accuracy.
- FR-3: Backend must expose REST API endpoints for single predictions and batch compliance runs.
- FR-4: Interface must include interactive metrics dashboard, analyst log table, and multi-step wizard.

2. Non-Functional Requirements

- NFR-1 (Aesthetics): Premium dark-mode glassmorphic theme with neon glows and smooth hover animations.
- NFR-2 (Latency): Endpoint response time must remain under 200ms.
- NFR-3 (Portability): App must support fully simulated offline operation if Python backend is disconnected.